

Conjecture 00000001673 is false:
the inducibility of C_6 is at most $2/7 < (\sqrt{3} - 1)/2$

Jackmeson1

The conjecture

Definition: the inducibility of C_6 is its maximal asymptotic density as an induced subgraph of n -vertex graphs. Conjecture: $\text{ind}(C_6) = (\sqrt{3} - 1)/2$; the extremal configuration is the balanced blow-up of the triangular prism, uniquely; and uniqueness is verified by the stabilizer structure of triangular-prism blow-ups.

We refute the value clause (A) for every graph on $n \geq 7$ vertices, and hence for the limit, the limsup and the liminf of the maximum induced density, and for the limsup along any sequence of graphs. We also refute the extremal-configuration clause (B): every blow-up of the triangular prism has no induced C_6 at all, while $\text{ind}(C_6) \geq 5/324 > 0$. The clause about the “stabilizer structure” is not a mathematical statement we formalize; it presupposes (B).

Definitions and reading

Definition 1 (Induced copy). For a graph G and a 6-set $S \subseteq V(G)$, S induces C_6 if S is the vertex set of an induced embedding of the 6-cycle, i.e. an injective $f: \mathbb{Z}/6 \rightarrow V(G)$ with $f(i)f(j) \in E(G) \iff j - i \equiv \pm 1 \pmod{6}$. Equivalently $G[S] \cong C_6$.

Definition 2 (Density and inducibility). For G on n vertices, $d(G) = \#\{S : |S| = 6, S \text{ induces } C_6\} / \binom{n}{6}$, and $M(n) = \max_{|V(G)|=n} d(G)$. Following Pippenger and Golumbic, as quoted by Hatami, Hirst and Norine (arXiv:1108.5699): “Let $[i_H(n)]$ denote the maximum induced density of $[H]$ in any graph on $[n]$ vertices. ... The asymptotic behavior of $[i_H(n)]$ is captured by the inducibility of $[H]$ defined as $[\lim_{n \rightarrow \infty} i_H(n)]$. It is straightforward to see that this limit always exists.” (Bracketed symbols were lost in text extraction.) The Lean development defines $\text{ind}(C_6) := \limsup_n M(n)$ and also treats $\liminf_n M(n)$, any limit of $M(n)$, and $\limsup_n d(G_n)$ for an arbitrary sequence (G_n) .

Definition 3 (Prism and blow-ups). The triangular prism is $K_3 \square K_2$ (“the Cartesian product of a cycle graph with a single edge”, Wikipedia, *Prism graph*; “Triangular prism graph – 6 vertices, 9 edges”). A blow-up of H along a map $p: V \rightarrow V(H)$ has $uv \in E \iff p(u)p(v) \in E(H)$; it is balanced when all fibres $p^{-1}(x)$ have equal size (Hatami–Hirst–Norine: “the copies of two vertices are adjacent if and only if the originals are. If every vertex is replaced with the same number of copies, then the resulting graph is called a balanced blow-up.”).

Conventions: graphs are simple and loopless; $\mathbb{N}$ includes 0; densities are real numbers; $\sqrt{3}$ is the positive root. No normalisation other than $\binom{n}{6}$ is used.

Proof of (A)

Lemma 4 (Facts about C_6). *C_6 is 2-regular, has no twins (for $i \neq j$ some $k \notin \{i, j\}$ is adjacent to exactly one of i, j), and is triangle-free.*

Lemma 5 (Pair lemma). *Let b be a vertex set and $x \neq y$ in b such that $b \setminus \{x\}$ and $b \setminus \{y\}$ both induce C_6 . Then every $w \in b \setminus \{x, y\}$ satisfies $w \sim x \iff w \sim y$.*

Proof. Let $S = b \setminus \{x, y\}$. In the copy $b \setminus \{y\} = S \cup \{x\}$ the vertex w has degree 2 = $|N(w) \cap S| + [w \sim x]$; in $b \setminus \{x\} = S \cup \{y\}$ it has degree 2 = $|N(w) \cap S| + [w \sim y]$. Hence $[w \sim x] = [w \sim y]$. $\square$

Lemma 6 (Local bound). *For every vertex set b , at most two $x \in b$ have $b \setminus \{x\}$ inducing C_6 .*

Proof. Suppose x, y, z are distinct and all three deletions induce C_6 . By the pair lemma applied to y, z , the vertices y, z have the same neighbours in $b \setminus \{y, z\}$. But y, z both lie in the induced copy $b \setminus \{x\}$, and since C_6 has no twins there is $w \in b \setminus \{x\}$, $w \neq y, z$, adjacent to exactly one of y, z . Contradiction. $\square$

Theorem 7 (Main bound). *For every $n \geq 7$ and every graph G on n vertices, $d(G) \leq 2/7$.*

Proof. Let c be the number of 6-sets inducing C_6 . Double count pairs (S, T) with S such a 6-set, $|T| = 7$, $S \subset T$. Each S lies in exactly $n - 6$ sets $T = S \cup \{x\}$; each T contains at most 2 such S (each $S \subset T$ is $T \setminus \{x\}$; local bound). So $c(n - 6) \leq 2\binom{n}{7}$. With $7\binom{n}{7} = (n - 6)\binom{n}{6}$ and $n - 6 > 0$ this gives $7c \leq 2\binom{n}{6}$. $\square$

Since $(11/7)^2 = 121/49 < 3$, $\sqrt{3} > 11/7$, so $(\sqrt{3} - 1)/2 > 2/7$. Therefore $M(n) \leq 2/7$ for $n \geq 7$, hence $\limsup M$, $\liminf M$, any limit of M , and $\limsup d(G_n)$ along any sequence are all $\leq 2/7 < (\sqrt{3} - 1)/2$. In particular (A) fails in every one of these readings.

Proof of (B)

Theorem 8. *No blow-up of the triangular prism (balanced or not) contains an induced C_6 .*

Proof. Let f be an induced embedding of C_6 into the blow-up along p , and $g = p \circ f$. If $g(i) = g(j)$ with $i \neq j$, then $f(i), f(j)$ have identical adjacency to every $f(k)$, so i, j are twins in C_6 , impossible. So g is injective into the 6 prism vertices, hence bijective, and $i \sim j \iff g(i)g(j) \in E(\text{prism})$. The prism triangle $(0, 0), (1, 0), (2, 0)$ then pulls back to a triangle in C_6 , impossible. $\square$

Theorem 9 (Lower bound). *For $m \geq 1$, $M(6m) \geq 5/324$; hence $\text{ind}(C_6) = \limsup M \geq 5/324$.*

Proof. In the blow-up of C_6 on $\{0, \dots, 6m - 1\}$ along $v \mapsto v \bmod 6$, each $\varphi: \{0..5\} \rightarrow \{0..m - 1\}$ gives the induced copy $i \mapsto 6\varphi(i) + i$, and distinct φ give distinct vertex sets. So $c \geq m^6$, while $\binom{6m}{6} \leq (6m)^6/6!$, giving $d \geq 720/6^6 = 5/324$. $\square$

Hence every sequence of prism blow-ups has density identically 0, which does not tend to $\text{ind}(C_6) \geq 5/324$, and at every $n = 6m$ the maximum density exceeds theirs by at least $5/324$.

Lean formalization

File `lean/Conjecture1673/Basic.lean` (Lean 4, Mathlib v4.33.1, only import `Mathlib`).
Key definitions: `C1673.InducesC6` (exists an embedding $f : \text{cycleGraph } 6 \rightarrow G$ of type `SimpleGraph.Embedding` with image S), `c6Count`, `c6Density`, `maxC6Density` (iSup over all graphs on $\text{Fin } n$), `indC6` ($\text{limsup } \text{maxC6Density } \text{atTop}$), `prism` ($K_3 \boxtimes K_2$ via `SimpleGraph.boxProd`), blow-ups as `prism.comap p`. Key theorems: `pair_twins`, `erase_count_le_two`, `count_mul_le`, `c6Density_le`, `two_sevenths_lt`, `indC6_le`, `liminf_le`, `lim_le`, `seq_limsup_le`, `prism_blowup_no_C6`, `prism_c6Density`, `indC6_ge`, and the main theorem `C1673.not_conjecture`, which states the conjunction of: $\text{density} < (\sqrt{3} - 1)/2$ for all $n \geq 7$; $\text{indC6} \neq (\sqrt{3} - 1)/2$; $\text{lim inf} \neq (\sqrt{3} - 1)/2$; $M(n) \not\rightarrow (\sqrt{3} - 1)/2$; for every graph sequence $\text{limsup } d(G_n) \neq (\sqrt{3} - 1)/2$; no sequence of prism blow-ups has density tending to `indC6`; prism blow-ups on $6m$ vertices have density at least $5/324$ below $M(6m)$.

Axiom audit: `#print axioms C1673.not_conjecture` reports `[propext, Classical.choice, Quot.sound]`; no `sorry`, no `native_decide`. The finite facts about C_6 are checked by kernel `decide` on `Fin 6`.